



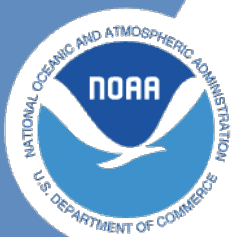
NOAA Technical Memorandum NMFS-XXX-##

My GrandOrgue Implementation

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January 2023

U.S. DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric
Administration
National Marine Fisheries Service
Northwest Fisheries Science Center



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Making sense of scattered documentation and user experiences

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Executive Summary

This site documents my experience setting up, configuring, and using a GrandOrgue virtual pipe organ (VPO). As of 2026, there was little compiled documentation other than how to download the software. Information such as system and hardware requirements was scattered throughout the internet, but one generally had to find it through a search engine. Some of this information was later discovered within the GrandOrgue Help pages, but that was after downloading, installing, and digging into the software. I wanted to know what I needed *before* starting the project to determine whether it was feasible. For example, could I use a Raspberry Pi or would I need something more powerful? Lots of community user experiences were captured in hundreds of issues and thousands of discussions on the official GitHub repo. It all felt very scattered. This site attempts to organize my research, trials, and errors.

These pages are my contribution to the open source GrandOrgue project and its outstanding user community. I hope these pages help you build and enjoy your own GrandOrgue virtual pipe organ.

1 Hardware

Requirements and considerations

1.1 BLUF

The minimum requirements needed to get started with GrandOrgue are:

- Computer with at least 8 GB RAM (16+ GB preferred)
- Touch screen monitor (for interacting with virtual console controls)
- MIDI-USB cable or MIDI cable with a USB audio interface device for connecting the computer to the MIDI instrument
- MIDI-ready organ or keyboard

And, if you don't want the sound to play from your computer speakers:

- Audio cable to port audio from computer to an organ amplifier
or
- Quality external speakers to connect directly to the computer

1.2 Discussion

GrandOrgue is free open source computer software supported for Windows, MacOS, and at least some Linux distributions. System and hardware requirements are summarized here, with the most important requirement being **RAM**.

My initial installation was intended to be a proof-of-concept learning experiment to assess whether or not it was something I wanted to delve further into and scale up. I

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already had an old (circa 1997) Casio MIDI keyboard¹, so I decided to use that as the workbench test case for experimentation. Then, once I knew what I was doing and felt comfortable with the technical components, I wanted to transfer the setup to a MIDI-ready digital church organ to try it out “in the field” with an actual organ console, decent speakers, and a room where an organ is regularly used.

With this in mind, I sought portability with minimal spatial footprint. Desktop computers took up too much space and are not easily portable, so they were out. Without having a personal organ at home, I needed to interface with the software directly through the computer rather than via an organ console, so I wanted to have a touch screen monitor that could be positioned in close proximity to the music keyboard. While standard laptops with touch screens are portable and could work, I would have to always reach over the laptop keyboard to reach the screen, which was not ideal. My solution was to use a **Raspberry Pi** together with a simple **touch screen monitor**.

I opted for the Raspberry Pi 5 because it was the first and, as of 2026, only unit to contain 16 GB of RAM. I purchased a full CanaKit starter kit for \$385 which included:

- Raspberry Pi 5 with 2.4Ghz 64-bit quad-core CPU and 64 GB of RAM
- 128GB Micro SD Card pre-loaded with 64-bit Raspberry Pi OS, USB MicroSD Card Reader
- Plastic protective case
- Low noise bearing system fan
- Heat sink
- 45W PD Power Supply
- 2 6-foot display cables

I bought a simple and compact 15.6” portable touch screen monitor for \$95 as my primary means of interacting with GrandOrgue. An upgrade might be in store down the road, but this will have to start.

Unfortunately, the Raspberry Pi 5 does not have a dedicated 3.5mm headphone audio jack like its predecessor. Audio is outputted through HDMI or USB instead. I ended up buying a Behringer UMC204HD **USB audio interface** for \$116 which would allow me to use a single USB port for both audio and to connect standard MIDI cables instead of a MIDI-USB cable. I also bought a 9.8-foot dual 5-pin **MIDI cable** for \$11 to go along with that because I did not have one already.

Altogether, my expenses for the initial set up came out to be just over \$600:

¹Old keyboards can almost always be found on Facebook Marketplace or similar sites for free or very little money. Just make sure it supports MIDI. Most manufactured after the 1980s will.

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Table 1.1: Total cost of my initial proof-of-concept setup.

Expense	Cost (USD)
Raspberry Pi 5 kit	\$385
Behringer UMC204HD	\$116
Touch screen monitor	\$95
MIDI cable	\$11
Keyboard	\$0
TOTAL	\$607

Not bad for an authentic-sounding practice organ! The pages that follow describe how I set up, configured, and tested everything.

2 Set-up

Setting up the hardware and software for GrandOrgue

2.1 Raspberry Pi setup

Raspberry Pi (RPi) devices run Debian-based Raspberry Pi OS. GrandOrgue needs the 64-bit Raspberry Pi OS in order to utilize all of the installed RAM (a 32-bit OS would cap the usable RAM at 4 GB, regardless of what is installed in the system.) The 64-bit OS conveniently comes pre-installed on the CanaKit RPi.

Before installing anything, download and install any OS updates:

```
sudo apt update && sudo apt full-upgrade -y
sudo reboot
```

GrandOrgue recommends X11 to properly handle window sizing and positioning. To switch the display mode:

1. Open a terminal and run `sudo raspi-config`
2. Use the down arrow key to select **Advanced Options**
3. In the next menu, select **Wayland**
4. Select **X11**, then **<OK>**
5. Use the right arrow key to select **<Finish>** and then **Reboot**

The touch monitor was plug-and-play. Since the RPi 5 only has a single USB-C port, which is used for the power supply, the monitor needed to be connected to the RPi via both HDMI for data and USB for power. I used the HDMI to mini-HDMI cable that came with the RPi and the USB-C to USB-A cable that came with the monitor. The RPi did need to be rebooted after everything was connected for the touch screen to work properly.

2.2 GrandOrgue installation

From the GrandOrgue releases GitHub, download the arm64.deb architecture package. As of May 2026, the current release was grandorgue_3.17.1-1_arm64.deb. You can also download it directly in the terminal:

```
wget https://github.com/GrandOrgue/grandorgue/releases/download/3.17.1-1/grandorgue_3.17.1-
```

(Note: grandorgue requires **wxWidgets 3.2** and grandorgue-wx30 requires **wxWidgets 3.0**. Raspberry Pi OS Bookworm ships with wxWidgets 3.2, so the standard grandorgue package above is the correct one.)

Install the software using either the downloaded installer or terminal:

```
sudo apt install ./grandorgue_3.17.1-1_arm64.deb
```

You might get a Permission Denied warning, but it can be ignored:

```
N: Download is performed unsandboxed as root as file '/home/username/grandorgue_3.17.1-1_ar
```

It is not a fatal error; the application should have installed fine.

Verify the installation by looking for it in the **Sound & Video** application menu. Launch the program. It should open and load the demo organ.

2.2.1 MIDI

Next, configure the MIDI hardware:

1. Plug in the Behringer UMC204HD USB audio interface to the RPi via USB and to the keyboard or organ using MIDI cables.
2. Open GrandOrgue and go to **File** ▢ **Preferences** ▢ **Audio**. You should see the UMC204HD listed as an ALSA device. Select it, set the sample rate to 48000 Hz and start with 512 samples per buffer.
3. You can tune the buffer size down later if latency feels sluggish, or up if you hear crackle.
4. Go to **File** ▢ **Preferences** ▢ **MIDI Devices**. The UMC204HD's MIDI ports should appear as ALSA MIDI devices.
5. Enable the input port. Then load a sample set.

2.2.2 GrandOrgue configuration

Next, we need to map MIDI channels between GrandOrgue and the instrument.

1. Go to **File** ▢ **Settings** ▢ **MIDI Devices**
2. Under **MIDI output devices**, make sure UNC204HD is selected. Click **OK**.
3. On the virtual pipe organ (VPO) console, right-click a virtual manual to map to.
4. In the box that opens, click **Listen for Event**. Play a key on the keyboard you wish to map to that virtual organ manual.
5. Click **OK**.

To reduce audio lag:

1. **File** ▢ **Settings** ▢ ((Audio))o
2. Set the **Samples per buffer** to 256 or 128 (defaults to 512).

The lower the buffer size, the lower the latency. At 48,000 Hz sample rate, 128 samples per buffer supposedly gives about 2.7 milliseconds of latency, which is completely imperceptible. The tradeoff is that very low buffer sizes can cause audio crackle or dropouts if the RPi can't process audio fast enough. The Pi 5 should, however, be plenty powerful enough to handle 128 samples without trouble, especially with the demo sample set.

Even at 128 samples/buffer, I still noticed a lag, but it was better than when I started. It is possible that some delay is "built in" as the result of the sample recording. This remains to be determined. I am accustomed to old-technology, non-realistic digital organs on which the sound is instantaneous, whereas many pipe organs have a slight (fraction of a second) lag between the time a key is depressed and the sound is actually heard.

2.2.3 Audio

By default, the VPO plays through the computer speakers – in this case, the speakers on the touch screen monitor. Much better sound quality can be obtained with better speakers or headphones.

Bluetooth headphones may experience a fairly significant lag in audio. It did for me. Since my old keyboard did not have an Audio In jack, I could not channel the sound back to the keyboard speakers – which was just as well, because they are not very good either. The best (but not perfect) option was wired headphones plugged into the UNC204HD interface. The only headphones I owned had a 1/8" plug, so I had to use a

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1/4"-1/8" adapter to connect them. There was a slight buzz, possibly resulting from the extra connection. It was annoying, but not bad enough to abandon the setup.

3 Customization

Using different organ sample sets

The default sample set is great to get started with, but it's rather small. The beauty of virtual pipe organ (VPO) software is the ability and flexibility to use any number of sample sets. Many sample sets of various sizes exist, from small single-division organs to large many-division instruments. One constraint is disk storage: one needs to have enough space to download and store the sample set. The primary constraint with choosing samples to use, however, is RAM: the sample set must fit in the computer's RAM to work properly. Thus, my constraint was 16 GB RAM that came on the Raspberry Pi 5. What this means in practice:

- **Under 8 GB:** loads comfortably, plenty of headroom
- **8–14 GB:** should work fine, some breathing room for OS
- **14–15 GB** getting risky; the OS and GrandOrgue itself need some RAM too, so you don't want to use all 16 GB for the sample set
- **Over 15 GB:** will not work; not enough memory for GrandOrgue and the OS

Since I only had one keyboard with no pedalboard, it made sense to consider single-manual samples. I could use larger ones, but I could only play one manual. The workaround would be mapping the Great division to my keyboard and relying on couplers to play other manuals from the Great manual.

3.1 Bureå Funeral Chapel

The first organ I tried was the Bureå Funeral Chapel from Lars Virtual Pipe Organs. This was a small chamber organ with one manual, pedals, and nine stops. All of the manual stops were split stops with separate drawknobs for bass (first two octaves) and discant. Split stops provide greater flexibility on a small instrument, allowing the organist to use different combinations of stops on one manual instead of across two manuals.

3 Customization

First, I downloaded the .rar file. I needed to install the “unrar” utility the first time in order to extract this:

```
sudo apt install unrar
```

Then I unpacked it into the GrandOrgue/Organs/ directory:

```
unrar x Burea_Funeral_Chapel.rar /home/username/GrandOrgue/Organs/Burea_Funeral_Chapel/
```

I then opened GrandOrgue, went to **File** → **Open**, found and selected the desired organ sample, and clicked **Open**. The new organ should load and be ready for use.

References